

National College of Ireland

Project Submission Sheet

Student Name: Jai Prabhas Malluri
Student ID: 24310875
Programme: MSc in Open Data Practice **Year:** 2026
Module: Research Practicum (MSCODP1)
Lecturer: Namita Agarwal
Submission Due Date: 13/02/2026
Project Title: Dynamic Personalization for AI Assistants: A Comparison between Fine-Tuning and Retrieval-Augmented Generation with Editable Memory Graphs
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Academic Integrity Certificate Verification Code: eIXCULydpA

I hereby certify that the information contained in this (my submission) is information pertaining to research I conducted for this project. All information other than my own contribution will be fully referenced and listed in the relevant bibliography section at the rear of the project.

ALL internet material must be referenced in the references section. Students are encouraged to use the Harvard Referencing Standard or appropriate referencing style supplied by the Library. To use other author's written or electronic work is illegal (plagiarism) and may result in disciplinary action. Students may be required to undergo a viva (oral examination) if there is suspicion about the validity of their submitted work.

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Signature: M Jai Prabhas

Date: 13/02/2026

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2. Projects should be submitted to your Programme Coordinator.
3. **You must ensure that you retain a HARD COPY of ALL projects**, both for your own reference and in case a project is lost or mislaid. It is not sufficient to keep a copy on computer. Please do not bind projects or place in covers unless specifically requested.
4. You must ensure that all projects are submitted to your Programme Coordinator on or before the required submission date. **Late submissions will incur penalties.**
5. All projects must be submitted and passed in order to successfully complete the year. **Any project/assignment not submitted will be marked as a fail.**

6. **Please check that you read AI and Academic Integrity Acknowledgement Supplements in this document**

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AI Acknowledgement Supplement

Research Practicum (MSCODP1)

Dynamic Personalization for AI Assistants: A Comparison between Fine-Tuning and Retrieval-Augmented Generation with Editable Memory Graphs

Your Name/Student Number	Course	Date
Jai Prabhas Malluri/24310875	MSc in Open Data Practice	13/02/2026

This section is a supplement to the main assignment, to be used if AI was used in any capacity in the creation of your assignment; if you have queries about how to do this, please contact your lecturer. For an example of how to fill these sections out, please click [here](#).

AI Acknowledgment

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool
NA	NA	NA

Description of AI Usage

This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how these responses were utilized or modified in the assignment. **One table should be used for each tool used.**

[Insert Tool Name]	
[Insert Description of use]	
[Insert Sample prompt]	[Insert Sample response]

Evidence of AI Usage

This section includes evidence of significant prompts and responses used or generated through the AI tool. It should provide a clear understanding of the extent to which the AI tool was used in the assignment. Evidence may be attached via screenshots or text.

Additional Evidence:

[Place evidence here]

Additional Evidence:

[Place evidence here]

Dynamic Personalization for AI Assistants: A Comparison between Fine-Tuning and Retrieval-Augmented Generation with Editable Memory Graphs

Jai Prabhas Malluri

24310875

Programme Code – Research Practicum CA1

National College of Ireland

1. Research Problem Background

Current Large Language Models (LLMs) have great conversational capabilities but struggle with the dynamic, long-term, personal user data (memories) of their users needed by highly-personalized AI assistants. Traditional model fine-tuning hard-wires this user data into the parametric memory of the model. However, this approach is computationally costly for updating real time, and also prone to hallucination or obsolescence when the states of the user, which are inevitable to change. An alternative is Retrieval-Augmented Generation (RAG), which provides models access to non-parametric, non-dynamic, memory-base external to the model on a dynamic basis. Managing personal memories have particular problem in editing, requiring constant insertion, deletion, and replacement of time-sensitive facts (e.g change flight schedules, expiring hotel bookings). Recent frameworks include Editable Memory Graph (EMG) combined with RAG which provide structured partitioned management of these memories. However, to prove this viability conclusively for a voice activated assistant (such as a "Jarvis" clone) this approach on RAG will have to be more rigorously benchmarked against purely fine-tuned architectures in a bid to prove enhancements in factual accurate and dynamic mutability of knowledge.

2. Research Question

- Is RAG with Editable Memory Graphs better than fine-tuning for providing AI assistant instant and accurate updates?
- Does RAG assist in hallucinations and does it update faster than fine-tuning for personalized voice AI?

3. Justification

This research question has a value as demand of highly personalized AI assistants grows and requires learning how to adapt to the daily changes of self of the users, without the cost of retraining cycles, computationally prohibitive, of fine tuning. The question becomes important because it directly impinges upon the key limitations of static parametric memory of modern LLMs because it suggests a more robust paradigm of human-computer interaction. It is feasible given there are open-source implementations of LLM frameworks, as well as techniques of testing EMG-RAG that are known like using Reinforcement Learning (RL) on Markov Decision Processes (MDP) for mem selection are readily available for implementation. Furthermore, the question is clear and measurable that the comparative performance of the generated answers can be quantitatively judged, using well-established Natural Language Processing metrics such as ROUGE, BLEU and Exact Match (EM) accuracy based on task e.g. question answering and autofill services. Finally, the approach is also ethical and privacy-conscious because by using a RAG-based graph system managed independent on a per-user basis, it avoids severe privacy risks of having sensitive personal data embedded into a fine-tuned model's weights on a global scale.

4. Specific Items to be Addressed

Data Collection and Structuring Pipeline: The creation of a secure and local pipeline for the extraction of raw personal user data (schedules, relationships, preferences etc) and structuring them into a multi-layered Editable Memory Graph composed of Relationship, Preference, Event and Attribute nodes.

Architecture Implementation: Development of voice to text and text to speech integration of an orchestrator for the voice artificial assistant along with implementing the Fine-tuned baseline model and an EMG-RAG pipeline using RL agent to traverse nodes in the graphs active by user queries.

Continuous Editability Testing: Testing of the system's ability to cope with dynamic personal information by testing the performance decay and adaptability of applying insertions, deletions and replacements to the memory graph over a simulated period of time.

Comparative Benchmarking: To quantitatively compare the conversational outputs both architectures with ground truth dynamic data, ROUGE (R -1/2/L), BLEU and Exact Match accuracy are used to determine the better approach in reduction of hallucination and improvement of factual recall.

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